

Aishwarya Aravind Nair

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Education

University of Wisconsin - Madison: Masters in Information | GPA : 3.8/4.0

Sept 2023 – May 2025

University of Mumbai Bachelor of Engineering, Instrumentation

Aug 2017 – June 2021

Skills

Certifications: Azure Administrator Associate (AZ-104)

Languages & Frameworks: Python, C, JavaScript, SQL, FastAPI

Cloud & Infrastructure: AWS (Lambda, EC2, EKS, S3, DynamoDB, CloudWatch), Azure (AKS, App Gateway, VMs), Terraform, Docker, Kubernetes, Helm

Data & Streaming: Kafka, Airflow, Snowflake, PostgreSQL, Redis, PySpark

DevOps & Monitoring: CI/CD (GitHub Actions, Azure DevOps), Prometheus, Grafana, ELK Stack

Distributed Systems: Microservices, REST APIs, Message Queues, Load Balancing, High Availability

Experience

DevOps Engineer Intern, XNODE Inc. – Remote

Oct 2025 – Present

- Built Helm-based CI/CD pipelines for AWS EKS that reduced deployment time from 15 to 6 minutes; implemented Prometheus and Grafana monitoring dashboards to track deployment health and alert on failures
- Deployed and configured Model Context Protocol (MCP) servers on Kubernetes, building custom REST API endpoints to expose tool capabilities for AI-workflow integration and client POCs
- Automated infrastructure provisioning using Terraform and AWS CDK, reducing manual setup time by 70% and ensuring consistent environment configuration across dev/staging/prod environments

Adaptive Technology Assistant, University of Wisconsin - Madison – Madison, WI

Oct 2023 – May 2025

- Built Python-based automation tools to convert 1000+ PDF documents to accessible formats (HTML, ePub) using OCR and document parsing libraries (PyPDF2, pdfplumber), reducing manual conversion time by 85%
- Provided assistive technology support to 500+ students, troubleshooting screen readers (JAWS, NVDA), magnification software, and voice recognition tools across Windows/macOS environments

Cloud Engineer, Assistant System Engineer, Tata Consultancy Services Ltd – Mumbai, India

Jun 2021 – Aug 2023

- Migrated manual deployment process to automated CI/CD pipeline using Terraform, Python, and Azure DevOps for fintech platform processing 50Cr+ monthly transactions, reducing release windows from 5 hours to 2 hours; developed reusable Terraform modules standardizing infrastructure provisioning across dev/staging/prod
- Architected and deployed Azure App Gateway with AKS (Kubernetes) infrastructure supporting 50K+ daily transactions, implementing SSL/TLS termination, rate limiting, and health monitoring that maintained 99.99% uptime during peak loads
- Executed zero-downtime MySQL migration from Single Server to Flexible Server for production database, achieving <5 minutes downtime through careful planning, replication setup, and automated failover scripts
- Engineered automated log archival solution using Python and Azure Blob Storage (Cool Tier), moving 100GB/month of MySQL logs and cutting storage costs by 40% while maintaining compliance requirements
- Built Azure Monitor dashboards and automated VM scheduling policies for non-production environments, reducing monthly cloud spend by 15% and improving operational visibility across 50+ resources

Projects

Task Management REST API | Python/FastAPI, PostgreSQL, Redis, Docker

2025

- Built RESTful API with JWT authentication, role-based access control, and request validation; containerized with Docker and documented with Swagger UI
- Designed normalized database schema with proper indexing; implemented Redis caching for frequently accessed resources and wrote unit/integration tests using pytest

Event-Driven Processing Pipeline | Kafka, Python, PostgreSQL, Docker

2025

- Built event streaming pipeline using Kafka with consumer groups for parallel processing; implemented dead letter queue for handling failed messages
- Designed idempotent consumers ensuring data consistency and fault tolerance; added Grafana dashboards tracking consumer lag, message throughput, and error rates